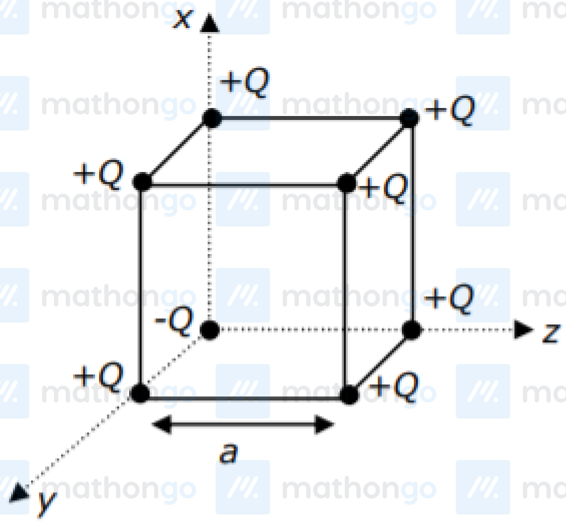
Q1: 24 Feb (Shift 1) - Single Correct

A cube of side 'a' has point charges +Q located at each of its vertices except at the origin where the charge is -Q. The electric field at the centre of cube is:



(1) $\frac{2Q}{3\sqrt{3}\pi\varepsilon_0 a^2}(\hat{x}+\hat{y}+\hat{z})$

(2) $\frac{\mathrm{Q}}{3\sqrt{3}\pi\varepsilon_0 \mathrm{a}^2}(\hat{\mathrm{x}}+\hat{\mathrm{y}}+\hat{\mathrm{z}})$

(3) $\frac{-2\mathrm{Q}}{3\sqrt{3}\pi\varepsilon_0 \mathrm{a}^2}(\hat{\mathrm{x}}+\hat{\mathrm{y}}+\hat{\mathrm{z}})$

(4) $\frac{-\mathrm{Q}}{3\sqrt{3}\pi\varepsilon_0 \mathrm{a}^2}(\hat{\mathrm{x}}+\hat{\mathrm{y}}+\hat{\mathrm{z}})$

Q2: 24 Feb (Shift 2) - Single Correct

Two electrons each are fixed at a distance '2d'. A third charge proton placed at the midpoint is displaced slightly by a distance $x(x << d)$ perpendicular to the line joining the two fixed charges. Proton will execute simple harmonic motion having angular frequency:

$(\mathrm{m} = \text{mass of charged particle})$

(1) $\left(\frac{q^2}{2\pi\varepsilon_0 m d^3}\right)^{\frac{1}{2}}$

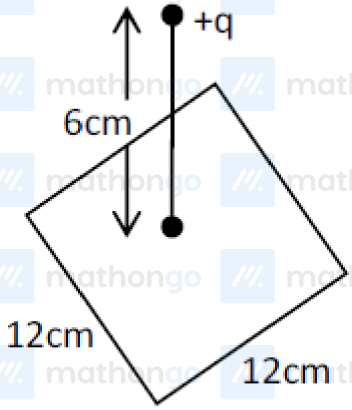
(2) $\left(\frac{\pi\varepsilon_0 \mathrm{md}^3}{2\mathrm{q}^2}\right)^{\frac{1}{2}}$

(3) $\left(\frac{2\pi\varepsilon_0 m d^3}{q^2}\right)^{\frac{1}{2}}$

(4) $\left(\frac{2q^2}{\pi\varepsilon_0 m d^3}\right)^{\frac{1}{2}}$

Q3: 24 Feb (Shift 2) - Numerical

A point charge of $+12 \mu C$ is at a distance 6 cm vertically above the centre of a square of side 12 cm as shown in figure. The magnitude of the electric flux through the square will be ____ $\times 10^{3} Nm^{2}/C$



Q4: 25 Feb (Shift 1) - Numerical

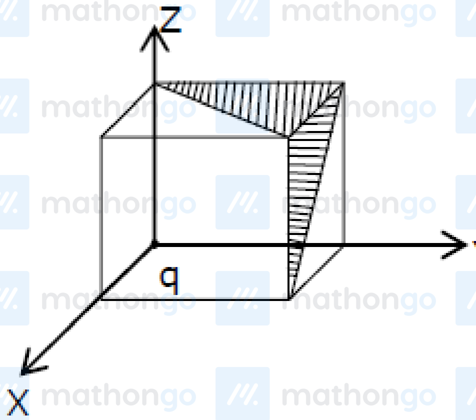
The electric field in a region is given by $\vec{E}=\left(\frac{3}{5} E_{0} \hat{\imath}+\frac{4}{5} E_{0} \hat{\jmath}\right) \frac{N}{c}$. The ratio of flux of reported field through the rectangular surface of area $0.2 \mathrm{~m}^{2}$ (parallel to $\mathrm{y}-\mathrm{z}$ plane) to that of the surface of area $0.3 \mathrm{~m}^{2}$ (parallel to x -z plane) is a : b, where a = ____ (round off to nearest integer)
[Here $\hat{\mathrm{i}}, \hat{\mathrm{j}}$ and $\hat{\mathrm{k}}$ are unit vectors along x, y and z-axes respectively]

Q5: 25 Feb (Shift 1) - Numerical

512 identical drops of mercury are charged to a potential of 2 V each. The drops are joined to form a single drop. The potential of this drop is ____ V.

Q6: 25 Feb (Shift 2) - Single Correct

A charge 'q' is placed at one corner of a cube as shown in figure. The flux of electrostatic field E though the shaded area is:



(1) $\frac{q}{48\varepsilon_0}$

(2) $\frac{q}{8\varepsilon_0}$

(3) $\frac{q}{24\varepsilon_0}$

(4) $\frac{q}{4\varepsilon_0}$

Q7: 25 Feb (Shift 2) - Numerical

Two small spheres each of mass 10mg are suspended from a point by threads 0.5 m long. They are equally charged and repel each other to a distance of 0.20 m. Then charge on each of the sphere is $\frac{a}{21} \times 10^{-8}$C. The value of 'a' will be

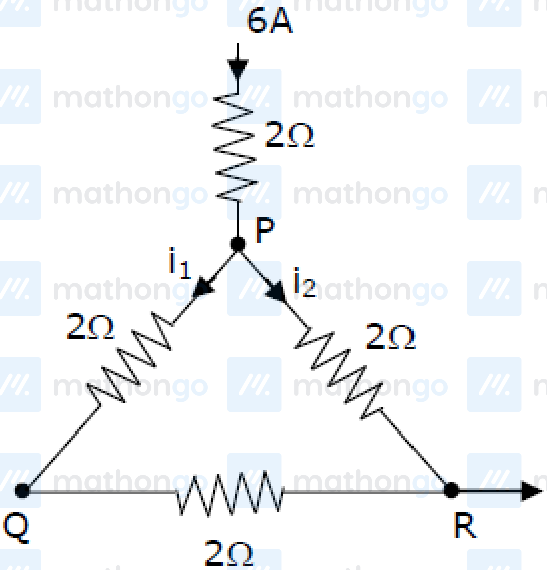
Q8: 25 Feb (Shift 2) - Numerical

Two identical conducting spheres with negligible volume have 2.1nC and -0.1nC charges, respectively. They are brought into contact and then separated by a distance of 0.5 m. The electrostatic force acting between the spheres is ____ $\times 10^{-9}$ N

[Given : $4\pi\varepsilon_0 = \frac{1}{9\times10^9}$ SI unit]

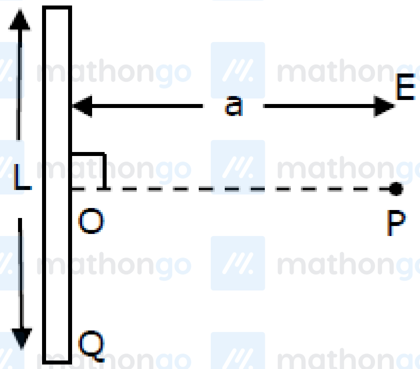
Q9: 25 Feb (Shift 2) - Numerical

A current of 6 A enters one corner P of an equilateral triangle PQR having 3 wires of resistance 2Ω each and leaves by the corner R. The currents i_1 in ampere is



Q10: 26 Feb (Shift 1) - Single Correct

Find the electric field at point P (as shown in figure) on the perpendicular bisector of a uniformly charged thin wire of length L carrying a charge Q. The distance of the point P from the centre of the rod is $a = \frac{\sqrt{3}}{2}L$



(1) $\frac{Q}{2\sqrt{3}\pi\varepsilon_0 L^2}$
(2) $\frac{\sqrt{3}Q}{4\pi\varepsilon_0 L^2}$
(3) $\frac{Q}{3\pi\varepsilon_0 L^2}$
(4) $\frac{Q}{4\pi\varepsilon_0 L^2}$

Q11: 26 Feb (Shift 1) - Numerical

In an electrical circuit, a battery is connected to pass 20C of charge through it in a certain given time. The potential difference between two plates of the battery is maintained at 15 V. The work done by the battery is ____ J.

Q12: 26 Feb (Shift 2) - Single Correct

Given below are two statements :

Statement - I : An electric dipole is placed at the centre of a hollow sphere. The flux of electric field through the sphere is zero but the electric field is not zero anywhere in the sphere.

Statement - II : If R is the radius of a solid metallic sphere and Q be the total charge on it. The electric field at any point on the spherical surface of radius $\mathrm{r}(<\mathrm{R})$ is zero but theelectric flux passing through this closed spherical surface of radius r is not zero.

In the light of the above statements. Choose the correct answerfrom the option given below :

Option :

(1) Statement I is true but Statement II is false

(2) Statement I is false but Statement II is true

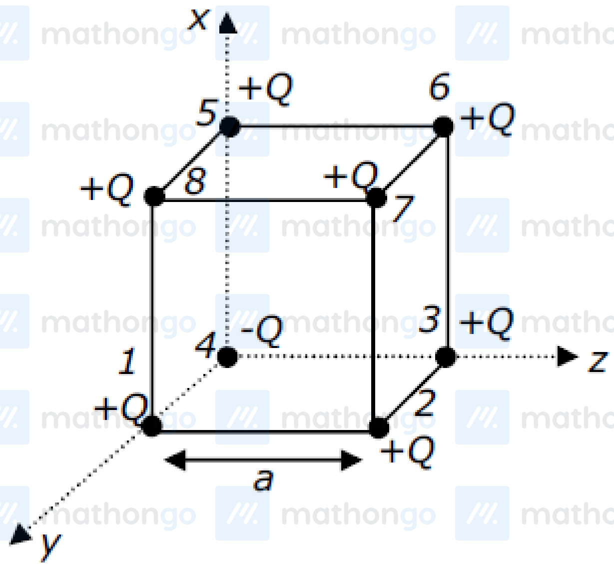
(3) Both Statement I and Statement II are true

(4) Both Statement I and Statement II are false

Answer Key

Q1 (3)	**Q2** (1)	**Q3** (226)	**Q4** (1)
Q5 (128)	**Q6** (3)	**Q7** (20)	**Q8** (36)
Q9 (2)	**Q10** (1)	**Q11** (300)	**Q12** (1)

Q1 (3)



If only +Q charges are placedat the corners of cube of side a then electric field at the centre of the cube will be zero.

But in the given condition one $(-\mathrm{Q})$ is placed at one corner of cube so here

$\mathrm{E}_1 = \mathrm{E}_6, \mathrm{E}_2 = \mathrm{E}_5$ and $\mathrm{E}_3 = \mathrm{E}_8$ (So it will cancel out each other so electric field at centre is due to Q_4 and Q_7

Here electric field at centre $= 2(\mathrm{E.f.})_4$

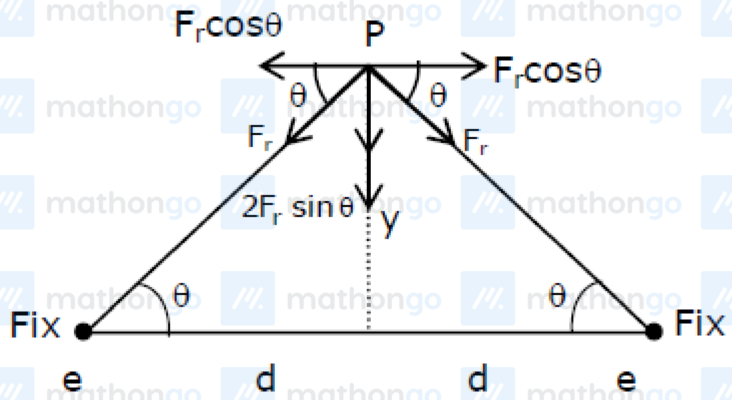
As, $|\mathrm{E}_4| = |\mathrm{E}_7|$

$(E.F)_C = \frac{2\mathrm{kQ}}{\left(\frac{\sqrt{3}\mathrm{a}}{2}\right)^2} = \frac{8\,\mathrm{K}Q}{3\mathrm{a}^2} \quad \left\{\because \mathrm{K} = \frac{1}{4\pi\varepsilon_0}\right\}$

$(E \cdot F)_C = \frac{2Q}{3a^2\pi\varepsilon_0}$

In vector form $\Rightarrow \vec{E} = \frac{-2Q}{3a^2\pi\varepsilon_0} \times \left(\frac{\hat{x}+\hat{y}+\hat{z}}{\sqrt{3}}\right)$

Q2 (1)



Restoring force on proton :-

$F_r = \frac{2Kq^2y}{[d^2+y^2]^{\frac{3}{2}}}$

$Y <<< d$

$F_r = \frac{2kq^2y}{d^3} = \frac{q^2y}{2\pi\varepsilon_0 d^3} = ky$

$\mathrm{K} = \frac{\mathrm{q}^2}{2\pi\varepsilon_0\ \mathrm{d}^3}$

Angular Frequency :$\omega = \sqrt{\frac{k}{m}}$

$\omega = \sqrt{\frac{q^2}{2\pi\varepsilon_0 m d^3}}$

Q3 (226)

Using Gauss law, it is a part of cube of side 12 cm and charge at centre so;

$\phi = \frac{\mathrm{Q}}{6\varepsilon_0} = \frac{12\mu\mathrm{c}}{6\varepsilon_0} = 2\times 4\pi \times 9 \times 10^9 \times 10^{-6}$

$= 226 \times 10^3 \mathrm{Nm}^2/\mathrm{C}$

Q4 (1)

$\phi = \vec{E}_1\vec{A}$

$\vec{A}_a = 0.2\hat{\imath}$

$\vec{A}_b = 0.3\hat{\jmath}$

$\phi_a = \left(\frac{3}{5}E_0\hat{\imath} + \frac{4}{5}E_0\hat{\jmath}\right)\cdot 0.2\hat{\imath}$

$\phi_a = \frac{3}{5}E_0 \times 0.2$

$\phi_a = \left(\frac{3}{5}E_0\hat{\imath} + \frac{4}{5}E_0\hat{\jmath}\right)\cdot 0.3\hat{\jmath}$

$\phi_b = \frac{4}{5}E_0 \times 0.3$

$\frac{a}{b} = \frac{\phi_a}{\phi_b} = \frac{\frac{3}{5}E_0 \times 0.2}{\frac{4}{5}E_0 \times 0.3} = \frac{6}{12} = 0.5$

Q5 (128)

Let charge on each drop $=$ q

radius $= r$

$v = \frac{kq}{r}$

$2 = \frac{\text{kq}}{\text{r}}$

radius of bigger

$\frac{4}{3}\pi R^3 = 512 \times \frac{4}{3}\pi r^3$

$R = 8r$

$v = \frac{k(512)q}{R} = \frac{512}{8}\frac{kq}{r} = \frac{512}{8} \times 2$

$= 128$ V

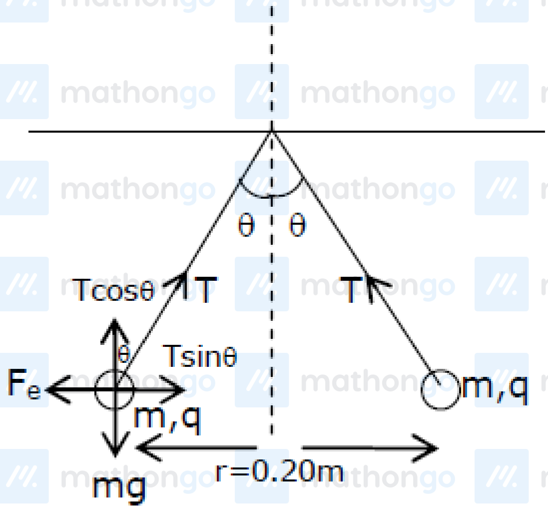
Q6 (3)

$\phi = \frac{q}{24\varepsilon_0}$

$\phi_T = \left(\frac{q}{24\varepsilon_0} + \frac{q}{24\varepsilon_0}\right) \times \frac{1}{2}$

$\phi_T = \frac{q}{24\varepsilon_0}$

Q7 (20)



$T\sin\theta = \frac{kq^2}{r^2}$

$T\cos\theta = mg$

$\tan\theta = \frac{\mathrm{kq}^2}{\mathrm{mgr}^2}$

$q^2 = \frac{\tan\theta mgr^2}{k}$

$\because \tan\theta = \frac{0.1}{0.5} = \frac{1}{5}$

$q^2 = \frac{1}{5} \times \frac{10\times10^{-6}\times10\times0.2\times0.2}{9\times10^9}$

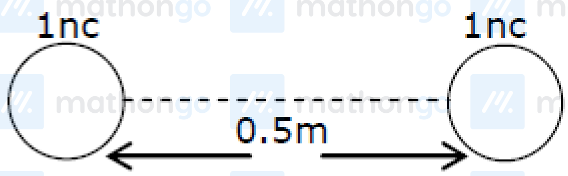
$q = \frac{2\sqrt{2}}{3} \times 10^{-8}$

after comparison from the given equation $\mathrm{a} = 20$

Q8 (36)

When they are brought into contact & then separated by a distance $= 0.5\ \mathrm{m}$

Then charge distribution will be



The electrostatic force acting b/w the sphere is

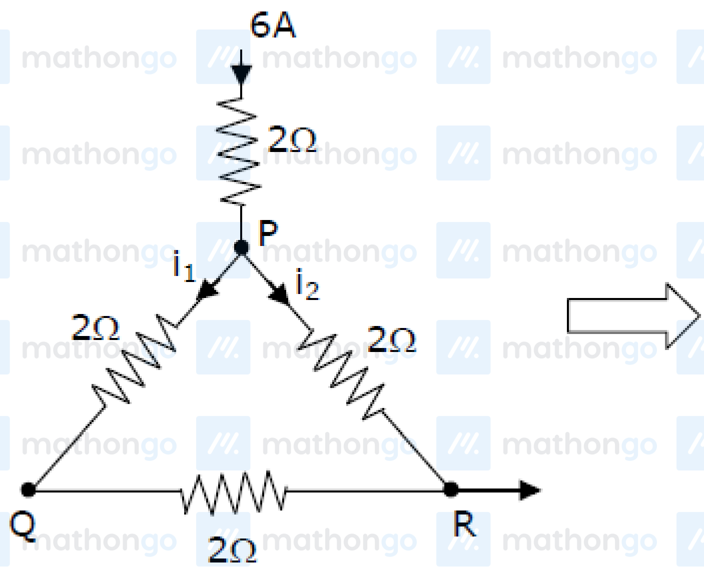
$F_e = \frac{kq_1q_2}{r^2}$

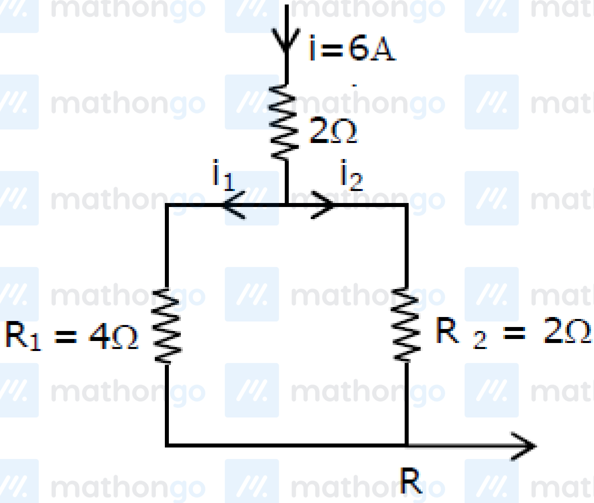
$= \frac{9\times10^9\times1\times10^{-9}\times1\times10^{-9}}{(0.5)^2}$

$= \frac{900}{25} \times 10^{-9}$

$F_e = 36 \times 10^{-9}$ N

Q9 (2)



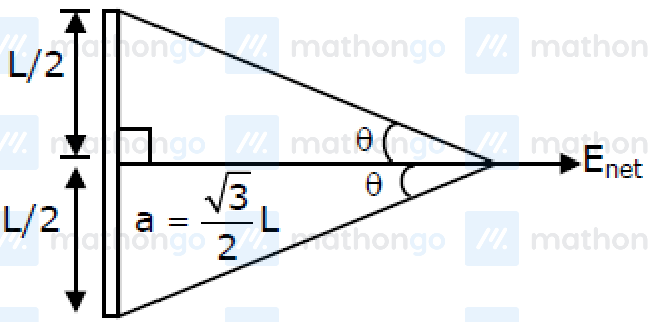


The current $i_1 = \left(\frac{R_2}{R_1+R_2}\right) i$

$= \left(\frac{2}{4+2}\right) \times 6$

$i_1 = 2\ A$

Q10 (1)



$\tan\theta = \frac{L/2}{\frac{\sqrt{3}}{2} L} \Rightarrow \frac{1}{\sqrt{3}}$

$\theta = 30^{\circ}$

$E_{net} = \frac{K\lambda}{\frac{\sqrt{3}}{2} L}(\sin 30^{\circ} + \sin 30^{\circ}) \Rightarrow \frac{2KQ}{\sqrt{3}\, L^2}\left(\frac{1}{2} + \frac{1}{2}\right)$

$E_{net} = \frac{1}{4\pi\varepsilon_0} \frac{2Q}{\sqrt{3}L^2}$

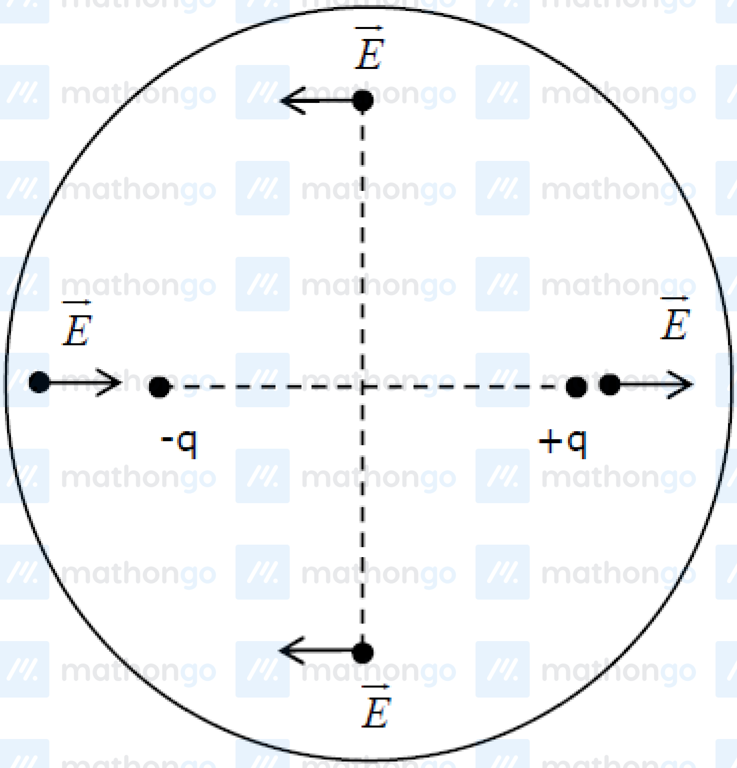
$E_{net} = \frac{Q}{2\sqrt{3}\pi\varepsilon_0\, L^2}$

Q11 (300)

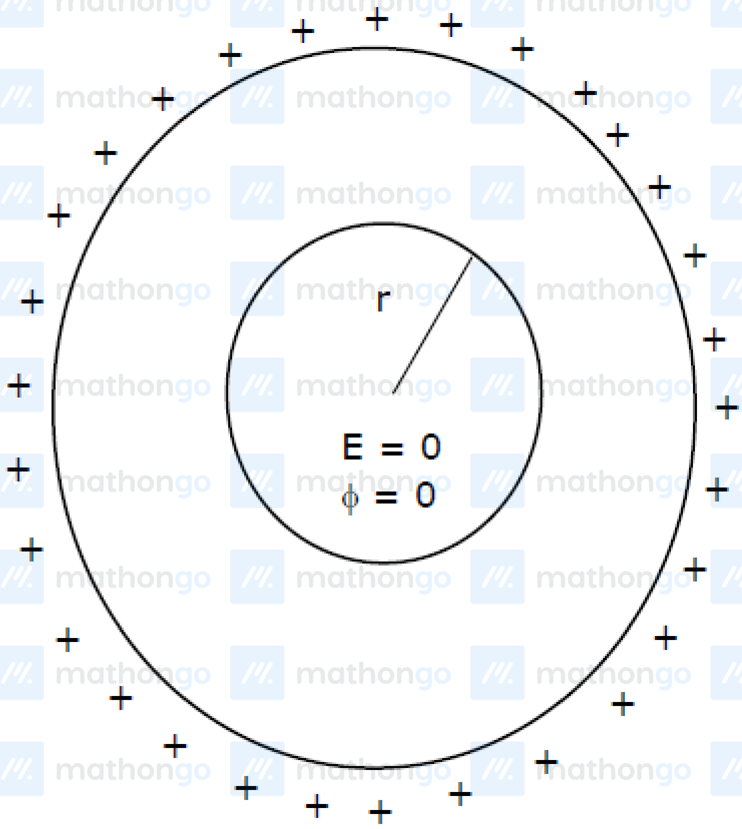
Charge flown $(Q) = 20c$ Potential difference $(\mathrm{V}) = 15\ \mathrm{V}$

$$\begin{aligned}\text{Work done } (w) &= Q.V \\ &= 20 \times 15 = 300\ \mathrm{J} \\ w &= 300\ \mathrm{J}\end{aligned}$$

Q12 (1)



Statement $-1 \rightarrow$ Correct



Statement $-2 \rightarrow$ Incorrect